wise been declared to be non-null. If all the columns of a foreign key are non-null in a given tuple, the usual definition of foreign-key constraints is used for that tuple. If any of the foreign-key columns is null, the tuple is defined automatically to satisfy the constraint.

This definition may not always be the right choice, so SQL also provides constructs that allow you to change the behavior with null values; we do not discuss the constructs here. To avoid such complexity, it is best to ensure that all columns of a **foreign key** specification are declared to be non-null.

Transactions may consist of several steps, and integrity constraints may be violated temporarily after one step, but a later step may remove the violation. For instance, suppose we have a relation *marriedperson* with primary key *name*, and an attribute *spouse*, and suppose that *spouse* is a foreign key on *marriedperson*. That is, the constraint says that the *spouse* attribute must contain a name that is present in the *person* table. Suppose we wish to note the fact that John and Mary are married to each other by inserting two tuples, one for John and one for Mary, in the above relation. The insertion of the first tuple would violate the foreign key constraint, regardless of which of the two tuples is inserted first. After the second tuple is inserted the foreign key constraint would hold again.

To handle such situations, integrity constraints are checked at the end of a transaction, and not at intermediate steps.[1]

6.3 Assertions

An **assertion** is a predicate expressing a condition that we wish the database always to satisfy. Domain constraints and referential-integrity constraints are special forms of assertions. We have paid substantial attention to these forms of assertion because they are easily tested and apply to a wide range of database applications. However, there are many constraints that we cannot express by using only these special forms. Two examples of such constraints are:

- The sum of all loan amounts for each branch must be less than the sum of all account balances at the branch.
- Every loan has at least one customer who maintains an account with a minimum balance of $1000.00.

An assertion in SQL takes the form

create assertion <assertion-name> **check** <predicate>

Here is how the two examples of constraints can be written. Since SQL does not provide a "for all X, $P(X)$" construct (where P is a predicate), we are forced to im-

1. We can work around the problem in the above example in another way, if the *spouse* attribute can be set to null: We set the spouse attributes to null when inserting the tuples for John and Mary, and we update them later. However, this technique is rather messy, and does not work if the attributes cannot be set to null.